



The elliptic relation between release angle and period for a pendulum

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(Received January 16, 2024, Published online February 5, 2024)

Two prominent models that give the period of a pendulum are the ideal pendulum, utilizing the small angle approximation, and the non-ideal pendulum, utilizing an elliptic integral. These models were compared against each other for pendulums of varying angles across 5 different pendulum lengths. While the ideal pendulum works well for small angles, its accuracy falls off rapidly as the release angle of the pendulum is increased. The elliptic model, while harder to work with, maintains its accuracy at both large and small angles.

I. INTRODUCTION AND HISTORY

Throughout the years, pendulums have been utilized by humanity's greatest minds to help us determine and understand how the world around us works. The pendulum was studied by many leading researchers in the seventeenth century, including Galileo, Hooke, Huygens, and Newton. Its motion was studied to ascertain the value of the acceleration due to gravity and its variance between polar regions to the equator. Pendulums have been used to establish collision and conservation laws, among many others. Thanks to the simple relation for a pendulum's period when swinging at small angles, they could then be utilized in clockwork, allowing timekeeping to be much more accurate than ever before. Innovations were made to further improve this accuracy. The measurements made possible by pendulums in clockwork advanced studies in mechanics, navigation, and astronomy. The pendulum and its applications were vital in bringing about the scientific revolution [1].

Beyond its research applications, the pendulum is a staple in physics classrooms, both at the university and pre-university levels. It consists merely of a weight attached to a length of string or a rod, and it is allowed to swing from a fixed point. The simplicity of the setup allows for easy access to pendulum-related experiments. Due to this simplicity, pendulum experiments can easily be set up remotely by individual students. This makes it a choice demo for online classes, something that is growing in demand [2].

Another aspect of the pendulum's simplicity allows it to be the base of many different experiments. The pendulum's versatility allows students to examine its behavior in many different circumstances. At small angles, the math describing the pendulum's motion is trivial, and perfect to help team beginning physics students. Curious students can then easily investigate how a pendulum behaves when the length, weight, and angle of the pendulum are changed. Combining pendulums can inform students on how coupled systems behave. Investigations

into driving and damping oscillations are easily achievable as well. The pendulum's simplicity allows it to be an accessible and versatile tool to assist with physics instruction and research.

II. ANALYSIS

When solving problems involving pendulums, it is common to assume the pendulum is ideal. This simplifies the calculation of the period T from a differential equation to the simple relation

$$T_0 = 2\pi\sqrt{\frac{l}{g}}, \quad (1)$$

where l is the length from the pendulum's pivot point to its center of mass, and g is the acceleration due to gravity. This relation comes from taking the small angle approximation to the differential equation of the pendulum's motion, but its accuracy relies on the release angle of the pendulum being small. As the angle increases, this model deviates from the motion a real pendulum exhibits. This differential equation can be solved without the small angle approximation; in doing so, we need an elliptic integral. This leaves us with the relation

$$T(\theta_{\max}) = T_0 \frac{2}{\pi} K\left(\sin\left(\frac{\theta_{\max}}{2}\right)\right) \quad (2)$$

for a non-ideal pendulum that swings at large angles. This model depends not only on the pendulum's length but also its release angle $\theta_{\max}$. $K(x)$ represents the complete elliptic integral of the first kind.

A lab was conducted to compare the accuracies of the ideal and non-ideal pendulum models for the determination of the period. To do this, a rod was attached to an angular motion sensor. Weights were affixed to the rod at a determined distance. The pendulum was released at angles varying from 0.01 rad to 1.58 rad. Five different pendulum lengths were tested: 31.5cm, 24.0cm, 16.5cm,

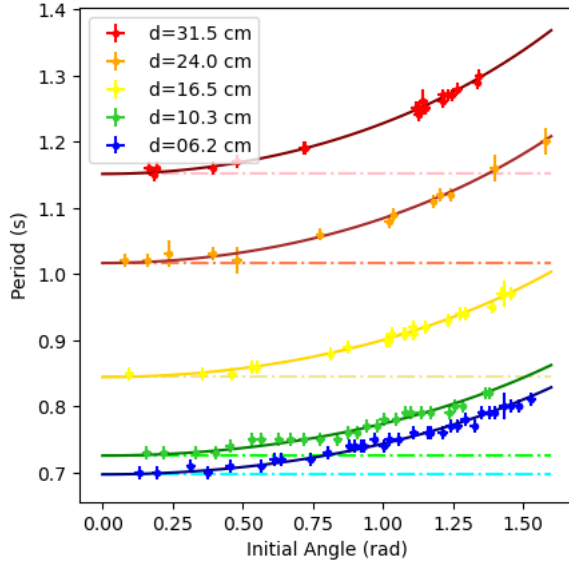


FIG. 1. Period plotted against initial angle for 5 different lengths of pendulum. The ideal pendulum relation can be seen in the dashed lines, while the elliptic model is represented by the solid lines. The elliptic model functions well for the spread of angles on each pendulum, but the ideal approximation is only accurate at angles less than 0.25 rad.

10.3cm, and 6.2cm. For each trial, the initial angle was measured, as well as the periods of the first and second oscillations. These were then plotted and fitted with the ideal and the non-ideal pendulum models. A correctional term b was included in the ideal approximation to ensure a better fit in both models, becoming

$$T_0 = 2\pi\sqrt{\frac{l+b}{g}}. \quad (3)$$

This investigation is in the nature of the model, not the specifics of it, so this correctional term b will be ignored. A χ^2 analysis was then taken for both models. The reduced χ^2 value for the elliptical models ranged from 0.1 to 0.3, indicating a strong fit between the model and the data. The ideal pendulum model betrays a χ^2 value between 20 to 90 as the release angle of the pendulum is increased. While the math is more difficult the elliptical model is much more accurate for predicting the period of a pendulum over a wide variety of angles.

III. CONCLUSION

Pendulums are simple machines, and at small angles, the math regarding them is simple. Unfortunately, as the maximum angle of an oscillation increases, the math regarding them grows in complexity. At small angles, the

period of a pendulum can be easily expressed in terms of the pendulum's length using the small angle approximation (Equation 1). As the angle increases, this model deviates from observed behavior in a real pendulum. To solve for the pendulum's motion without assuming small angles, an elliptic integral must be used, as in Equation 2. The math becomes considerably more complicated and difficult to do by hand, but it can easily be solved with the use of computers to accurately predict the behavior of a pendulum even at large angles.

To put these models to the test, an experiment was run to measure the periods of different pendulums released at varying angles, both large and small. The ideal and elliptic models were then applied to each set of data to determine the overall fit to the observed period. The elliptic model had much lower χ^2 values than the ideal model, indicating that it is more accurate over the varying angles and pendulum lengths in this setup. Considerations for future work include investigation into the correctional b term (Equation 3) as well as into the tendency toward increasing χ^2 values for the elliptic model as the pendulum length decreases.

Appendix A: Uncertainty

Uncertainty for the measured period was found by measuring the time of the first and second complete oscillations. The difference between these periods was then taken as the uncertainty. This method was chosen to account for the decay of a pendulum due to resistive forces. If the uncertainty was less than 0.01s, 0.01s was taken as the uncertainty as that was the time step between measurements.

Uncertainty for the release angle was taken as the smallest deviation the rotary motion detector could detect (0.017 rad).

Reduced χ^2 value was calculated using the formula

$$\chi^2 = \frac{\sum \frac{(P_{\text{obs}} - P_{\text{exp}})^2}{\sigma_\theta}}{\sqrt{(N)}}. \quad (A1)$$

Pendulum length (cm)	Ideal model χ^2	Elliptical model χ^2
31.5	85.23	0.19
24.0	42.46	0.14
16.5	46.83	0.13
10.3	24.36	0.30
06.2	37.76	0.24

TABLE I. Reduced χ^2 values for the ideal and non-ideal models.

- [1] M. R. Matthews, C. F. Gauld, and A. Stinner, *The pendulum: Scientific, historical, philosophical and educational perspectives* (Springer Science & Business Media, 2005).
- [2] M. Escobar, H. Fernandes, O. Allard, and Y. Erazo, in *2019 5th Experiment International Conference (exp.at'19)* (2019) pp. 388–393.